

SAI YASWANTH YENGISETTY

PostgreSQL DBA | AWS Cloud | Performance Tuning & High Availability

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PROFESSIONAL SUMMARY:

PostgreSQL Database Administrator with 2+ years of experience managing production-grade databases in AWS environments, specializing in performance tuning, high availability, and cloud database architecture. Familiar with DevOps concepts and tools including infrastructure automation and CI/CD workflows, with exposure gained while supporting PostgreSQL environments in AWS.. Proven ability to support high-concurrency systems, optimize query performance, and contribute to maintaining highly available database environments for data-intensive applications.

TECHNICAL SKILLS:

Databases:	PostgreSQL, AWS Aurora PostgreSQL, Amazon RDS PostgreSQL, Oracle
Database Administration:	Performance Tuning, Query Optimization, Indexing Strategies, Table Partitioning, VACUUM & ANALYZE, REINDEX, Backup & Recovery (PITR), WAL Management, Replication, High Availability (Patroni), Disaster Recovery (pgBackRest, Barman)
Cloud Platforms:	Amazon Web Services (AWS) – RDS, Aurora, EC2, CloudWatch, IAM, S3
DevOps Exposure:	Basic knowledge of Terraform, Ansible, Git, and CI/CD concepts with hands-on exposure while supporting database deployments and automation activities.
Operating Systems:	Linux – Ubuntu, Amazon Linux
Monitoring & Database Tools:	pgBadger, Amazon CloudWatch, pgAdmin4, DBeaver, Patroni, pgBackRest, Barman
Database Security:	Role-Based Access Control (RBAC), User & Role Management, Access Control Policies, Audit Support
Scripting & Automation:	Python (database automation, monitoring, log parsing, reporting), Bash
AI / Data Systems Exposure:	Text-to-SQL Systems, AI Data Pipelines, Chatbot Database Backends, Embedding-based Query Systems

PROFESSIONAL EXPERIENCE:

Billion Bright Solutions — PostgreSQL DBA | Remote | September 2024 – Present

Project 1: AI-Powered Chatbot for Customer Support

- Deployed and managed PostgreSQL on AWS EC2 instances with high availability using replication and failover strategies.
- Configured WAL archiving and implemented Point-in-Time Recovery (PITR) using pgBackRest.
- Implemented indexing, partitioning, and query optimization, reducing response latency by **38%**.
- Tuned PostgreSQL parameters (work_mem, shared_buffers, autovacuum) based on workload patterns.
- Set up automated backups, retention policies, and disaster recovery strategies.
- Monitored database performance using CloudWatch and pgBadger, identifying slow queries and bottlenecks.
- Automated DBA tasks (health checks, log analysis, alerts) using Python scripts.
- Investigated and resolved slow queries, locking issues, and high CPU utilization in production environments.
- Collaborated closely with **AI/ML teams** to optimize database access patterns during **model inference**.

Business Impact & OKRs

- Achieved 99.99% uptime using replication, failover strategies, and cross-zone deployment.
- Ensured data durability and zero data loss through automated backup retention and PITR setup.
- Reduced query execution time significantly, improving chatbot response performance.

Project 2: AI Professional Notification Generator for RupeeEcom Apps

- Administered PostgreSQL clusters on AWS EC2 for a multi-tenant AI notification system.
- Designed schema isolation strategies for multi-tenant applications.
- Designed and managed streaming replication architecture (primary-standby) to ensure high availability and read scalability.
- Optimized queries, indexes, and materialized views for real-time notifications.
- Handled database incidents, replication lag issues, and performance bottlenecks.
- Implemented backup validation and recovery testing using pgBackRest.

Business Impact & OKRs

- Improved database throughput by **42%** through optimized **indexing and caching strategies**.
- Ensured zero downtime during peak traffic (sales/festivals).
- Enabled seamless scaling of the notification system during **festival and sale seasons**, ensuring **zero downtime**

Bytehubble.AI — PostgreSQL DBA Intern - Hyderabad, India | September 2023 – July 2024

Project: MegatuneDB — Natural Language to SQL Engine

Project Overview: Developed a **production-grade Text-to-SQL AI system** converting natural language into accurate SQL using a fine-tuned GPT-2 model, backed by PostgreSQL and exposed via a Streamlit interface.

Key Responsibilities

- Built a **Text-to-SQL system** that converts natural language queries into SQL queries.
- Used **PostgreSQL as the backend database** for log storage and analysis.
- Designed **schema-aware prompts and embeddings** for PostgreSQL tables with **24+ columns**.
- Developed a **Streamlit-based UI** enabling non-technical users to query databases conversationally.
- Used **Python for backend processing**, embeddings handling, and query orchestration.

PostgreSQL Expertise:

- Strong understanding of MVCC, WAL internals, checkpoints, and query planner.
- Experience in analyzing execution plans (EXPLAIN ANALYZE) for query optimization.
- Hands-on experience with autovacuum tuning and bloat management.
- Knowledge of replication slots, failover handling, and HA architecture.

CERTIFICATIONS:

- Generative AI Specialization — **Edureka**
- AWS Cloud Practitioner Essentials — **Coursera**
- Python for Data Science and Machine Learning — **Edureka**

EDUCATION:

Bachelor of Technology (B.Tech) in Computer Science and Engineering
Annamacharya Institute of Technology & Sciences, Tirupati, India
2021 – 2024 | CGPA: 7.77

LANGUAGES:

English | Telugu | Hindi

INTERESTS & PASSIONS:

Database performance engineering, cloud database architectures, high availability systems, and integrating AI-powered applications with scalable PostgreSQL backends.